

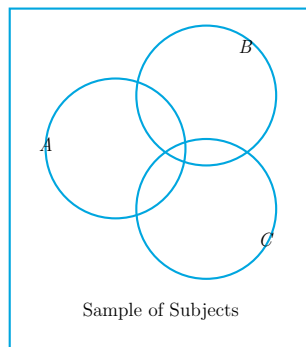
9.3.34.

A study was done to determine the efficacy of three different drugs—A, B, and C—in relieving headache pain. Over the period covered by the study, 50 subjects were given the chance to use all three drugs. The following results were obtained:

- 21 reported relief from drug A
- 21 reported relief from drug B
- 31 reported relief from drug C
- 9 reported relief from both drugs A and B
- 14 reported relief from both drugs A and C
- 15 reported relief from both drugs B and C
- 41 reported relief from at least one of the drugs.

Note that some of the 21 subjects who reported relief from drug A may also have reported relief from drugs B or C. A similar occurrence may be true for the other data.

- a. How many people got relief from none of the drugs?
- b. How many people got relief from all three drugs?
- c. Let A be the set of all subjects who got relief from drug A, B the set of all subjects who got relief from drug B, and C the set of all subjects who got relief from drug C. Fill in the numbers for all eight regions of the following diagram.



- d. How many subjects got relief from A only?
- a. How many people got relief from none of the drugs?

Let S be the set of all subjects in the study, and we know there were 50 subjects in total. Therefore:

$$N(S) = 50$$

Then, let A , B , and C be smaller sets inside of S :

$A = \{\text{subjects who got relief from drug A}\}$

$B = \{\text{subjects who got relief from drug B}\}$

$C = \{\text{subjects who got relief from drug C}\}$

We are looking for the subjects which got relief from none of the drugs. We know that 41 reported relief from at least one of the drugs, hence:

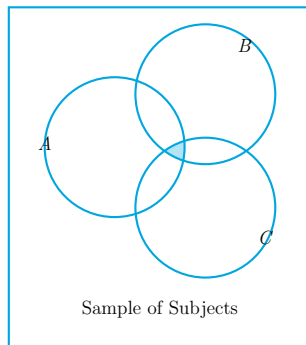
$$N(A \cup B \cup C) = 41$$

$$N(S) - N(A \cup B \cup C) = 50 - 41 = 9$$

Solution: a. There were 9 people who got relief from none of the drugs.

b. How many people got relief from all three drugs?

We are looking for the subjects which got relief from all three drugs, as exemplified by the colored region of the diagram:



We know that 41 reported relief from at least one of the drugs, hence:

$$N(A \cup B \cup C) = 41$$

We also know that 9 reported relief from both drugs A and B, 14 reported relief from both drugs A and C, and 15 reported relief from both drugs B and C. Hence:

$$N(A \cap B) = 9$$

$$N(A \cap C) = 14$$

$$N(B \cap C) = 15$$

And since the sets are finite, we can use the inclusion/exclusion rule:

$$\begin{aligned} N(A \cup B \cup C) = & N(A) + N(B) + N(C) - \\ & N(A \cap B) - \\ & N(A \cap C) - \\ & N(B \cap C) + \\ & N(A \cap B \cap C). \end{aligned}$$

So, we plug in our known numbers for our sets:

$$41 = 21 + 21 + 31 - 9 - 14 - 15 + N(A \cap B \cap C)$$

$$41 = 35 + N(A \cap B \cap C)$$

$$N(A \cap B \cap C) = 6$$

Solution: b. There were 6 people who got relief from all three drugs.

c. Let A be the set of all subjects who got relief from drug A, B the set of all subjects who got relief from drug B, and C the set of all subjects who got relief from drug C. Fill in the numbers for all eight regions of the following diagram.

Let's specify what we do know:

$$N(S) = 50$$

$$N(A) = 21$$

$$N(B) = 21$$

$$N(C) = 31$$

$$N(A \cap B) = 9$$

$$N(A \cap C) = 14$$

$$N(B \cap C) = 15$$

$$N(A \cup B \cup C) = 41$$

$$N(A \cap B \cap C) = 6$$

$$N(S) - N(A \cup B \cup C) = 9$$

Let's work from the center outward. We know $N(A \cap B) = 9$, but those 9 include the 6 in the middle. Similarly, we know $N(A \cap C) = 14$, and $N(B \cap C) = 15$, which also include the 6 in the middle. Hence:

$$\begin{aligned}
N(A \cap B) - N(A \cap B \cap C) &= 9 - 6 = 3 \\
N(A \cap C) - N(A \cap B \cap C) &= 14 - 6 = 8 \\
N(B \cap C) - N(A \cap B \cap C) &= 15 - 6 = 9
\end{aligned}$$

Now we are ready to calculate for A , B and C only:

$$A \text{ only} + A \cap B + A \cap C + A \cap B \cap C = 21 \Rightarrow A = 21 - 3 - 8 - 6 = 4$$

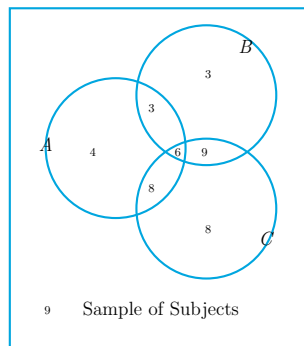
$$B \text{ only} + A \cap B + B \cap C + A \cap B \cap C = 21 \Rightarrow B = 21 - 3 - 9 - 6 = 3$$

$$C \text{ only} + A \cap C + B \cap C + A \cap B \cap C = 31 \Rightarrow C = 31 - 8 - 9 - 6 = 8$$

Now we can list our known values:

1. $A \text{ only} = 4$
2. $B \text{ only} = 3$
3. $C \text{ only} = 8$
4. $A \cap B$ only, meaning **not** $C = 3$
5. $A \cap C$ only, meaning **not** $B = 8$
6. $B \cap C$ only, meaning **not** $A = 9$
7. $A \cap B \cap C = 6$
8. Outside all three circles, meaning **none of the drugs** = 9

And plug them into the diagram:



And since the seven regions inside the circle give $4 + 3 + 8 + 3 + 8 + 9 + 6 = 41$, which agrees with $N(A \cup B \cup C) = 41$, then the 9 outside gives $41 + 9 = 50$, which is the total number of subjects.

Solution: c. See diagram.

d. How many subjects got relief from A only?

$$A \text{ only} + A \cap B + A \cap C + A \cap B \cap C = 21$$

$$A = 21 - 3 - 8 - 6 = 4$$

Solution: d. There were 4 people who got relief from drug A only.